

Question 1: Describe what happens during the fetch-execute cycle.

- Data and instructions are **fetch**ed from memory, RAM, or primary storage.
- Data and instructions are **stored using the registers** (e.g., a specific register storing an address or data).
- Data and instructions are **dec**oded (often split into opcode and operand).
- Data and instructions are **exec**uted or processed.
- The **ALU** performs logical and arithmetic calculations during the execution phase.

Answer:

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Question 2 (i): Identify component(s) matching the statements.

- Stores a single address:
- Stores frequently used instructions:
- Is a register:
- Stores all currently running data and instructions:

i. The table has **five** components of a computer, and **four** statements.

Tick (✓) **one or more** boxes in each row to identify which component(s) each statement describes.

Statement	MAR	MDR	Cache	Program Counter	RAM
It stores a single address					
It stores frequently used instructions					
It is a register					
It stores all currently running data and instructions					

[4]

Question 2 (ii): Identify one register not given in part (i) and its purpose.

- Register:
- Purpose:

Register

Question 3: Provide the name or definition for CPU components.

- **Program Counter (PC):**
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- **Control Unit (CU):**
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- **Memory Address Register (MAR):**
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Arithmetic Logic Unit (ALU): Performs the mathematical and logical calculations.

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Question 4: Reason why cache size affects CPU performance.

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Question 5: Why is the cruise control an embedded system?

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Question 6: Why a 3.2 GHz processor is faster than a 1.2 GHz processor.

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Question 7 (i): Why a computer **needs both primary and secondary storage.**

Question 7 (ii): Example of secondary storage and data stored.

Question 7 (iii): Correcting Virtual Memory (VM) statements.

Question 8 (a)(i): Define the term pixel.

Question 8 (a)(ii): Total file size of **10 images (800 x 500, **8-bit**) in **kilobytes**.**

Question 8 (a)(iii): Minimum bits needed for 240 colors.

Question 8 (b): Choosing storage type for moving files.

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Question 9 (a): Base Conversions.

9(a). Each row in the table contains a number that is written as a denary value, 8-bit binary value and a 2-digit hexadecimal value. Some values are missing.

Complete the table by writing in the missing value for each number.

Denary	8-bit binary	2-digit hexadecimal
38		26
	01001110	4E

Denary	8-bit binary	2-digit hexadecimal
156	10011100	
215		D7

[4]

Question 9 (b): Binary addition of 01110011 and 00110100.

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Question 9 (c): 2-place right binary shift on 01110111.

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Question 10: Character Sets description.

- A character set stores _____ characters. Each is given a _____ binary code. Upper and lower-case letters are given _____ codes. ASCII uses _____ per character. If 'F' is 70, then 'L' is _____.

Question 11 (a): Convert Hex 2F to denary.

Question 11 (b): Convert Binary 10110000 to Hex.

- Answer:

Question 12 (i): Identify correct sound sampling description.

Question 12 (ii): Effect of changing bit depth.

Question 13 (i): Two benefits of compressing data for email.

Question 13 (ii): Why lossy compression is inappropriate for text files.
